# Lancer1911 ASR Live

Offline Real-Time Speech Recognition

*离线实时语音识别 · LLM 矫正 · 多语言翻译*

Installation & User Guide v3.0 · 安装与使用说明

## 1. System Requirements 系统要求

| Item<br>*项目* | Requirement<br>*要求* |
|---|---|
| Hardware<br>*硬件* | Apple Silicon Mac (M1 / M2 / M3 or later)<br>*Apple Silicon Mac（M1 / M2 / M3 及以上）* |
| Memory<br>*内存* | 24 GB minimum, 48 GB recommended<br>*最低 24GB，推荐 48GB* |
| macOS<br>*系统* | 13 Ventura or later<br>*macOS 13 Ventura 或更高* |
| Storage<br>*存储空间* | ~11 GB for AI models<br>*AI 模型约 11GB* |
| Internet<br>*网络* | Only needed for first-time model download<br>*仅首次下载模型需要联网* |

⚠️ Lancer1911 ASR Live runs entirely on-device. Your voice data never leaves your Mac.
⚠️ *Lancer1911 ASR Live 完全在本地运行，语音数据不会上传到任何服务器。*

## 2. Installation 安装步骤

### 2.1 Install Homebrew 安装 Homebrew（已安装可跳过）

Open Terminal and run:
*打开终端，运行以下命令：*

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

### 2.2 Install ffmpeg 安装 ffmpeg

Required for MP3 recording:
*录音保存为 MP3 格式需要此工具：*

```
brew install ffmpeg
```

### 2.3 Set Up Python Environment 配置 Python 虚拟环境

Run these commands once in Terminal:
*在终端依次运行以下命令（只需执行一次）：*

```
# Create virtual environment
python3 -m venv ~/asr-env

# Activate it
source ~/asr-env/bin/activate

# Install dependencies
pip install mlx-whisper mlx-lm mlx-audio silero-vad sounddevice \
            fastapi "uvicorn[standard]" pywebview onnxruntime \
            pyannote.audio torch omegaconf
```

> You only need to run these steps once. The virtual environment persists across reboots.
> *以上步骤只需执行一次，虚拟环境重启后依然有效。*

## 3. Download AI Models 下载 AI 模型

Lancer1911 ASR Live requires two AI models: a speech recognition model (Whisper) and a language correction model (Qwen3). Download them once — they are cached locally and work offline forever after.
*Lancer1911 ASR Live 需要两个 AI 模型：语音识别模型（Whisper）和语义矫正模型（Qwen3）。下载一次后永久保存在本地，之后完全离线使用。*

#### Speech Recognition Model — Whisper 语音识别模型

| Model<br>*模型* | Size<br>*大小* | Speed<br>*速度* | Accuracy<br>*精度* | Recommended for<br>*适用场景* |
|---|---|---|---|---|
| whisper-large-v3-turbo | ~3 GB<br>*约 3GB* | Fast<br>*快* | Excellent<br>*优秀* | Daily use ★ Recommended<br>*日常使用 ★ 推荐* |

| | | | | |
|---|---|---|---|---|
| whisper-large-v3-mlx | ~6 GB<br>*约 6GB* | Slower<br>*稍慢* | Highest<br>*最高* | Maximum accuracy<br>*追求最高精度* |

```
source ~/asr-env/bin/activate

# Recommended
hf download mlx-community/whisper-large-v3-turbo

# Optional — highest accuracy
hf download mlx-community/whisper-large-v3-mlx
```

### Speech Recognition Model (Alternative) — SenseVoice 备选 ASR 模型

| Model<br>*模型* | Size<br>*大小* | Speed<br>*速度* | Recommended for<br>*适用场景* |
|---|---|---|---|
| SenseVoiceSmall | ~0.5 GB<br>*约0.5GB* | Fast<br>*较快* | Optional; 3-5x faster, emotion/event detection<br>*可选：速度3–5倍，支持情绪/事件检测* |

```
source ~/asr-env/bin/activate
pip install mlx-audio

hf download mlx-community/SenseVoiceSmall
```

### Correction & Translation Model — Qwen3 矫正与翻译模型

| Model<br>*模型* | Size<br>*大小* | Recommended for<br>*适用场景* |
|---|---|---|
| Qwen3-14B-4bit | ~8 GB<br>*约 8GB* | 48 GB RAM ★ Recommended<br>*48GB 内存 ★ 推荐* |
| Qwen3-8B-4bit | ~5 GB<br>*约 5GB* | 24 GB RAM<br>*24GB 内存* |
| Qwen3-30B-A3B-4bit | ~16 GB<br>*约16GB* | ≥48 GB RAM (optional)<br>*≥48GB 内存（可选）* |

```
# 48 GB RAM — recommended
hf download mlx-community/Qwen3-14B-4bit

# 24 GB RAM
hf download mlx-community/Qwen3-8B-4bit

# ≥48 GB RAM — optional, higher quality

# Method 1: HuggingFace
# hf download mlx-community/Qwen3-30B-A3B-Instruct-2507-4bit
```

```
# Method 2: ModelScope (推荐国内用户 / recommended for users in China)
pip install -U modelscope
mkdir -p ~/.cache/modelscope/hub/models--mlx-community--Qwen3-30B-A3B-Instruct-
2507-4bit

python3 - <<'PY'
from modelscope import snapshot_download
from pathlib import Path
local_dir = Path.home() / ".cache" / "modelscope" / "hub" / "models--mlx-
community--Qwen3-30B-A3B-Instruct-2507-4bit"
model_dir = snapshot_download(
    model_id="mlx-community/Qwen3-30B-A3B-Instruct-2507-4bit",
    local_dir=str(local_dir)
)
print("Downloaded to:", model_dir)
PY
```

> If downloads are slow or failing, try the mirror: export HF_ENDPOINT=https://hf-mirror.com
> *下载缓慢或失败时，可使用镜像：export HF_ENDPOINT=https://hf-mirror.com*

### Speaker Diarization Models (Optional) 发言人识别模型（可选）

When installed, the app automatically identifies up to 4 speakers and labels each subtitle card. Skip this section if speaker identification is not needed.

*安装后应用可自动识别最多 4 位发言人，并在字幕卡片上标注。不需要此功能可跳过本节。*

### Step 1 — Create a HuggingFace Read Token 第一步：创建 HuggingFace Read Token

Go to https://huggingface.co/settings/tokens and create a token of type Read (not fine-grained). Copy the token string.

*前往 https://huggingface.co/settings/tokens，创建类型为 Read（非 fine-grained）的 Token，复制 Token 字符串。*

### Step 2 — Accept Model Agreements 第二步：同意模型协议

Visit each model page while logged in and click “Agree and access repository”:

```
https://huggingface.co/pyannote/embedding
https://huggingface.co/pyannote/segmentation-3.0
```

*登录 HuggingFace 网页，分别进入以上两个模型页面，点击「Agree and access repository」。*

### Step 3 — Login and Download 第三步：登录并下载

```
source ~/asr-env/bin/activate
hf auth login   # paste your Read token when prompted
hf download pyannote/embedding
hf download pyannote/segmentation-3.0
```

> After installation the app detects the models automatically at startup. The sidebar will show “Speaker ID: On” when active. No restart required after download.
> *安装完成后应用启动时自动检测，左侧边栏显示「发言人识别：开启」即生效，无需重启。*

# 4. Install Lancer1911 ASR Live.app 安装应用

## 4.1 Copy to Applications 拷贝到应用程序文件夹

Open the DMG file and drag Lancer1911 ASR Live into the Applications folder.
*打开 DMG 文件，将 Lancer1911 ASR Live 拖入 Applications（应用程序）文件夹。*

## 4.2 Bypass Gatekeeper 绕过安全限制

Because Lancer1911 ASR Live is not notarized by Apple, macOS may block it on first launch.
*由于 Lancer1911 ASR Live 未经 Apple 公证，首次打开可能被系统拦截。*

Option A — Right-click method:
*方法 A — 右键方式：*

1. Right-click Lancer1911 ASR Live in Applications → Open
2. Click Open in the dialog that appears

Option B — Terminal:
*方法 B — 终端命令：*

```
xattr -cr /Applications/Lancer1911\ ASR\ Live.app
```

## 4.3 Grant Microphone Access 授权麦克风

On first launch, macOS will request microphone permission. Click Allow.
*首次启动时系统会请求麦克风权限，点击「允许」。*

If no prompt appears: System Settings → Privacy & Security → Microphone → enable Lancer1911 ASR Live.
*若未弹出请求：系统设置 → 隐私与安全性 → 麦克风 → 开启 Lancer1911 ASR Live。*

# 5. Using Lancer1911 ASR Live 使用说明

## 5.1 First Launch 首次启动

Allow 30–60 seconds on first launch for the AI models to load into memory. The status bar will show Ready when the app is fully loaded.
*首次启动需要 30–60 秒加载 AI 模型，顶栏显示「就绪」后即可使用。*

> Subsequent launches are faster because the models stay cached in system memory.
> *后续启动更快，因为模型已缓存在系统内存中。*

## 5.2 Start Transcribing 开始识别

- Click Start in the left sidebar or the top bar
  - *点击左侧「开始识别」或顶栏「● 开始」按钮*
- Speak naturally — pause briefly (≈0.6 s) at the end of each sentence
  - *正常说话，每句结束后稍停顿（约 0.6 秒）系统自动识别*
- Subtitles appear in real time: original text on top, translations below
  - *字幕实时显示：原文在上，译文在下*
- Click Stop to end the session and save the recording (if enabled)
  - *点击「停止识别」结束本次录音*

## 5.3 Interface Overview 界面说明

| Area<br>*区域* | Function<br>*功能* |
|---|---|
| Top bar<br>*顶栏* | Status, Export, Clear, Theme toggle, Settings<br>*状态、导出、清空、主题切换、设置* |
| Left sidebar<br>*左侧边栏* | Recording controls, language selection, VAD parameters, live metrics<br>*录音控制、语言选择、VAD 参数、实时指标* |
| Subtitle area<br>*字幕区域* | Real-time subtitles with waveform display<br>*实时字幕与波形显示* |
| Bottom bar<br>*底部搜索栏* | Keyword search with highlight & clear button, jump-to-latest<br>*关键词搜索高亮、清除按钮、跳转最新* |

## 5.4 Settings 设置说明

### ASR & LLM Models ASR 与 LLM 模型

The app automatically scans locally downloaded models. Select from the dropdowns — no manual path entry required.
*应用自动扫描本地已下载模型，直接从下拉菜单选择，无需手动填写路径。*

### Scene & Terminology 场景与术语

Enter domain-specific context and vocabulary to boost recognition accuracy for technical terms. This text is passed to both Whisper (as initial_prompt) and Qwen3 (as system context).
*输入领域描述和专业词汇，显著提升专业术语识别精度。该文本同时传递给 Whisper（initial_prompt）和 Qwen3（system context）。*

```
Example / 示例:
Cardiology consultation. Terms: atrial fibrillation, LVEF, CABG, PCI.
心血管科会诊。术语：心房颤动、左心室射血分数、冠状动脉旁路移植术。
```

### Translation Target Languages 翻译目标语言

Select one or more output languages. When you speak a selected language, it is automatically excluded from translations to avoid duplication.
*勾选一种或多种输出语言。说某语言时，该语言自动从译文中排除，避免重复显示。*

### Audio Input Parameters 音频输入参数

| Parameter<br>*参数* | Default<br>*默认值* | Notes<br>*说明* |
|---|---|---|
| Save Recording<br>*录音保存* | On / 开启 | Prompts native save dialog on stop<br>*停止时弹出原生保存对话框* |
| MP3 Bitrate<br>*MP3 码率* | 192 kbps | Options: 64 / 128 / 192 / 320 kbps<br>*可选：64 / 128 / 192 / 320kbps* |
| Silence threshold<br>*句尾静音阈值* | 0.6 s | Pause duration to trigger recognition<br>*停顿多久后触发识别* |
| VAD sensitivity<br>*VAD 灵敏度* | 0.45 | Higher = less sensitive; noisy env: 0.6–0.7<br>*越高越不敏感；嘈杂环境建议 0.6–0.7* |
| Max sentence length<br>*最长单句时长* | 20 s | Force-cuts after this duration<br>*超时强制切断* |
| Microphone<br>*麦克风设备* | System default<br>*系统默认* | Click ↻ to rescan devices<br>*点 ↻ 刷新已连接设备* |

## 5.5 Export 导出

Click Export ▾ in the top bar. Choose a language filter and format, then a native save dialog appears.
*点击顶栏「导出 ▾」，选择语言筛选和格式，弹出原生保存面板。*

| Format<br>*格式* | Description<br>*说明* |
|---|---|
| TXT<br>*纯文本* | Plain text with timestamps<br>*每条含时间戳的纯文本* |
| SRT<br>*字幕文件* | Standard subtitle format for video editors<br>*标准字幕格式，可导入视频软件* |
| JSON<br>*结构化数据* | Full metadata: timestamps, latency, all translations<br>*含时间戳、时滞、原文和所有译文* |
| Markdown<br>*Markdown* | Paste into Obsidian, Notion, etc.<br>*适合粘贴到 Obsidian、Notion 等笔记工具* |

Language filter: All (mixed), Original text (corrected), Original text (raw ASR), or any single language. When a single language is selected, both original-language entries and translated entries are included.
*语言筛选：全部混合、仅原文（矫正后）、仅原文（ASR 原始），或任意单一语言。选择单一语言时，作为原文或译文的条目均会包含。*

## 6. Troubleshooting 常见问题

| Issue<br>*问题* | Solution<br>*解决方法* |
|---|---|
| Window shows "Connecting" for a long time<br>*窗口长时间显示「连接中」* | Wait 30–60 s for models to load. The status bar shows Ready when done.<br>*等待 30–60 秒，顶栏显示「就绪」后即可操作。* |
| No translations appear<br>*没有翻译输出* | Open Settings, check target languages, click Save Settings.<br>*打开设置，勾选翻译目标语言并保存。若无效，删除 ~/.asrlive_settings.json 后重启。* |
| Repeated words (e.g. "nope nope nope…")<br>*出现大量重复词* | Whisper hallucination — built-in filter handles most cases. Try raising VAD sensitivity to 0.6–0.7.<br>*Whisper 幻觉，程序已内置过滤。可调高 VAD 灵敏度至 0.6–0.7。* |
| LLM correction is slow (>2 s)<br>*LLM 矫正时间过长* | Switch to Qwen3-8B-4bit in Settings, or reduce the number of target languages. |

| | |
|---|---|
| | *在设置中换用 Qwen3-8B-4bit，或减少翻译目标语言数量。* |
| Recording not saved<br>*录音没有保存* | Ensure ffmpeg is installed (brew install ffmpeg) and recording is enabled in Settings.<br>*确认已安装 ffmpeg（brew install ffmpeg），且设置中录音开关已开启。* |
| Microphone not working<br>*麦克风无法使用* | System Settings → Privacy & Security → Microphone → enable Lancer1911 ASR Live.<br>*系统设置 → 隐私与安全性 → 麦克风 → 开启 Lancer1911 ASR Live。* |
| No module named 'onnxruntime'<br>*缺少 onnxruntime 模块* | Run: pip install onnxruntime<br>*运行：pip install onnxruntime* |
| Speaker ID shows “Not installed”<br>*发言人识别显示「未安装」* | Follow Section 3.3. Use the “Check Speaker ID Setup” button in Settings for guided instructions.<br>*参照第 3.3 节安装。也可点击设置中的「检查发言人识别功能」按钮查看引导说明。* |